Name:_____

Section:_____

Quiz 4 **(Show your work using clear notation for each question)**

1. A technician has 6 CPUs: 4 CPUs have 4 cores each, and 2 CPUs have 8 cores each. Two CPUs are selected at random without replacement. Let X be the total number of cores in the two selected CPUs.

(a) (8 pts) Find the probability mass function (pmf) of X.

2pts

x	8	12	16
$f(x)$	$\frac{2}{5}$	$\frac{8}{15}$	$\frac{1}{15}$

$P(X=8) = \frac{\binom{4}{2}}{\binom{6}{2}} = \frac{6}{15} = \frac{2}{5}$ 2pts

[Selection of x – 2pts each $f(x)$ – 2pts]

$P(X=12) = \frac{\binom{4}{1}\binom{2}{1}}{\binom{6}{2}} = \frac{8}{15}$ 2pts

$P(X=16) = \frac{\binom{2}{2}}{\binom{6}{2}} = \frac{1}{15}$ 2pts

(b) (3 pts) Find the probability that the total number of cores is greater than 8.

$\underbrace{P(X>8)}_{1\text{ pt}} = \frac{8}{15} + \frac{1}{15} = \frac{9}{15} = \boxed{\frac{3}{5}}$ 2pts

[If they calculate pmf wrong but the method is correct for this question, give 2 pts in total]

2. (3 pts) Calculate $\sum_{x=0}^{51}\binom{51}{x} 0.5^x 0.6^{51-x}$.

$= (0.5+0.6)^{51}$ – 2pts

$= \boxed{129.1299}$ – 1 pts

3. (6 pts) The probability that Alex hits a target on a single attempt is 0.3. Find the probability that Alex gets his 3rd hit on the 6th attempt.

$X \sim NB(r=3, p=0.3)$ – 2pts [NB is 1 pt, parameters 1 pt]

$\underbrace{P(X=6)}_{1\text{ pt}} = \underbrace{\binom{5}{2} 0.3^3 (1-0.3)^3}_{2\text{pts}} = \boxed{0.09261}$ 1pt

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Quiz 4 **(Show your work using clear notation for each question)**

1. (6 pts) The probability that Jamie makes a basket is 0.25. Find the probability that Jamie makes the 2nd basket on the 5th shot.

$X \sim NB(r=2,\ p=0.25)$

$P(X=5) = \binom{4}{1} 0.25^2 (1-0.25)^3 = \boxed{0.1055}$

2. (3 pts) Calculate $\sum_{x=0}^{41} \binom{41}{x} 0.5^x 0.7^{41-x}$.

$= (0.5+0.7)^{41}$

$= \boxed{1763.7259}$

3.A lab has 7 batteries: 5 are standard batteries and 2 are high-capacity batteries. A standard battery provides 3 units of power, and a high-capacity battery provides 7 units. Two batteries are selected at random without replacement. Let X be the total power units from the selected batteries.

(a) (8 pts) Find the probability mass function (pmf) of X.

x	6	10	14
$f(x)$	$\frac{10}{21}$	$\frac{10}{21}$	$\frac{1}{21}$

$P(X=6) = \frac{\binom{5}{2}}{\binom{7}{2}} = \frac{10}{21}$

$P(X=10) = \frac{\binom{5}{1}\binom{2}{1}}{\binom{7}{2}} = \frac{10}{21}$

$P(X=14) = \frac{\binom{2}{2}}{\binom{7}{2}} = \frac{1}{21}$

(b) (3 pts) Find the probability that the total power units are greater than 10.

$P(X>10) = P(14) = \frac{1}{21}$

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Quiz 4 **(Show your work using clear notation for each question)**

1. A warehouse stores 6 machines: 3 are type A and 3 are type B. A type A machine produces 5 units, and a type B machine produces 9 units. Two machines are randomly selected without replacement. Let X be the total units produced.

(a) (8 pts) Find the probability mass function (pmf) of X.

x	10	14	18
$f(x)$	$\frac{1}{5}$	$\frac{3}{5}$	$\frac{1}{5}$

$$P(X=10) = \frac{\binom{3}{2}}{\binom{6}{2}} = \frac{3}{15} = \frac{1}{5}$$

$$P(X=14) = \frac{\binom{3}{1}\binom{3}{1}}{\binom{6}{2}} = \frac{9}{15} = \frac{3}{5}$$

$$P(X=18) = \frac{\binom{3}{2}}{\binom{6}{2}} = \frac{3}{15} = \frac{1}{5}$$

(b) (3 pts) Find the probability that the total units are greater than 10.

$$P(X>10) = P(X=14) + P(X=18)$$
$$= \boxed{\frac{4}{5}}$$

2. (6 pts) The probability that Morgan solves a puzzle correctly is 0.4. Find the probability that Morgan gets the 4th correct solution on the 7th attempt.

$$X \sim NB(r=4,\ p=0.4)$$

$$P(X=7) = \binom{6}{3} 0.4^4 (1-0.4)^3 = \boxed{0.1106}$$

3. (3 pts) Calculate $\sum_{x=0}^{45} \binom{45}{x} 0.7^x 0.4^{45-x}$.

$$= (0.7+0.4)^{45}$$
$$= \boxed{72.8905}$$

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Quiz 4 **(Show your work using clear notation for each question)**

1. (6 pts) The probability that Riley passes a checkpoint is 0.20. Find the probability that Riley gets the 3rd pass on the 6th attempt.

$X \sim NB(r=3, p=0.20)$

$P(X=6) = \binom{5}{2} 0.2^3 (1-0.2)^3$

$= \boxed{0.0410}$

2. (3 pts) Calculate $\sum_{x=0}^{45} \binom{45}{x} 0.8^x 0.5^{45-x}$.

$= (0.8+0.5)^{45}$

$= \boxed{134106.8167}$

3. A technician has 8 sensors: 6 small sensors and 2 large sensors. A small sensor records 2 signals, and a large sensor records 6 signals. Two sensors are selected without replacement. Let X be the total number of signals recorded.

(a) (8 pts) Find the probability mass function (pmf) of X.

x	4	8	12
$f(x)$	$\frac{15}{28}$	$\frac{3}{7}$	$\frac{1}{28}$

$P(X=4) = \frac{\binom{6}{2}}{\binom{8}{2}} = \frac{15}{28}$

$P(X=8) = \frac{\binom{6}{1}\binom{2}{1}}{\binom{8}{2}} = \frac{12}{28} = \frac{3}{7}$

$P(X=12) = \frac{\binom{2}{2}}{\binom{8}{2}} = \frac{1}{28}$

(b) (3 pts) Find the probability that the total number of signals is greater than 4.

$P(X>4) = P(X=8) + P(X=12)$

$= \boxed{\frac{13}{28}}$